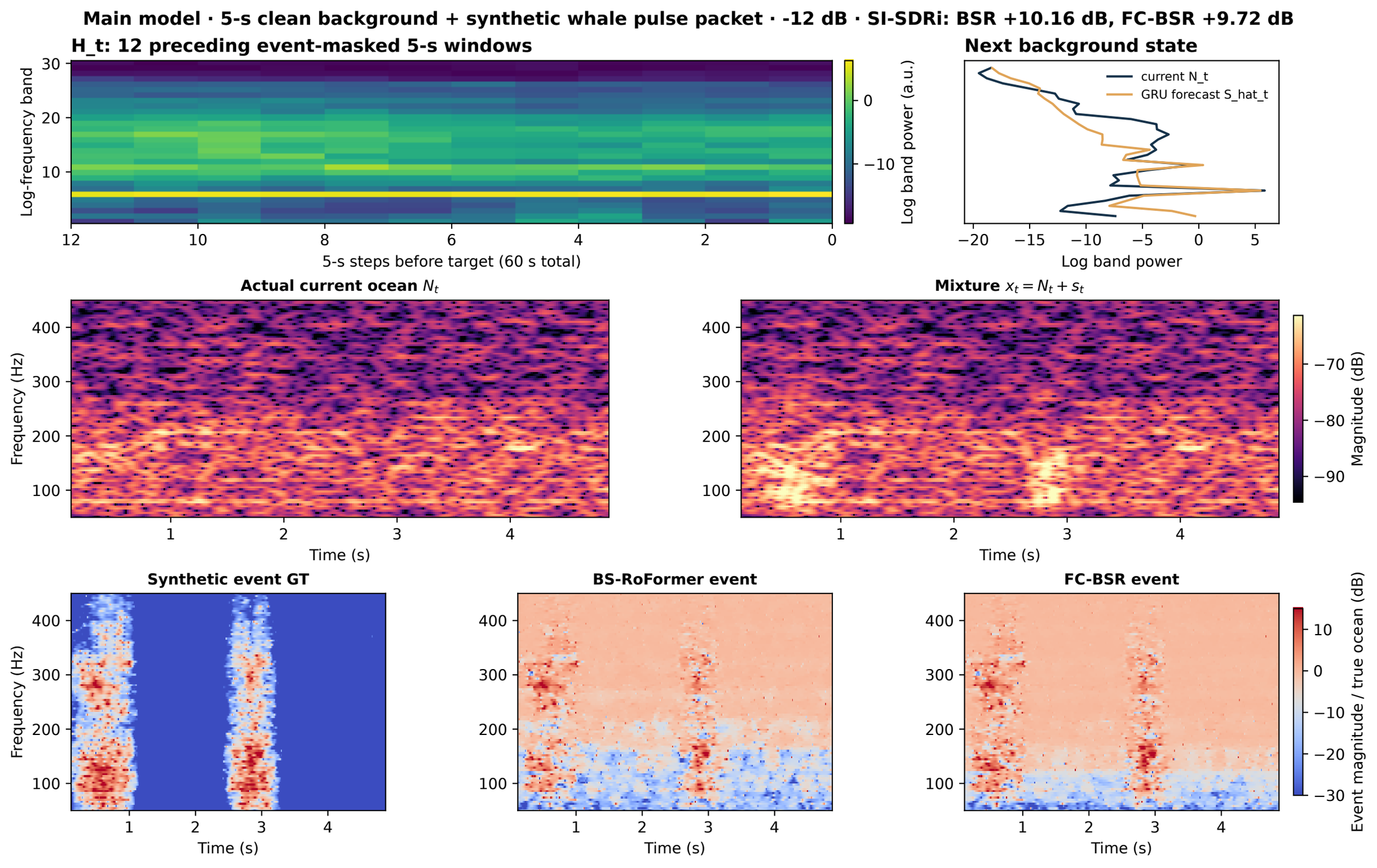

Main model · 5-s clean background + synthetic whale pulse packet · -12 dB · SI-SDRi: BSR +10.16 dB, FC-BSR +9.72 dB
H_t: 12 preceding event-masked 5-s windows
Log-frequency band
5-s steps before target (60 s total)
Next background state
current N_t
GRU forecast S_hat_t
Log band power (a.u.)
Log band power
Actual current ocean N_t
Mixture $x_t = N_t + s_t$
Frequency (Hz)
Time (s)
Magnitude (dB)
Synthetic event GT
BS-RoFormer event
FC-BSR event
Event magnitude / true ocean (dB)